

Maël Le Guillouzic

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Professional Summary

Master's student in AI and Applied Mathematics specializing in the application of deep learning to physical systems. Research experience in AI-driven ocean forecasting and atmospheric modeling. Seeking graduate programs bridging deep learning and computational physics.

Education

Télécom Paris | *Paris, France*

2022 – Present

- Master's in AI, Applied Mathematics, and Computer Science
- Coursework: Deep Learning, Image & Signal Processing, Language Theory, Data Science, Statistics
- #2 Engineering School in France (2023 ranking)

Preparatory Classes for Grandes Écoles (CPGE) | *Janson-de-Sailly, Paris*

2019 – 2022

- Intensive two-year program in advanced mathematics, physics, and engineering sciences
- Top 4% nationwide ranking in competitive entrance examinations

Professional Experience

Incoming Research Intern

Starting March 2026

CNRS – École Polytechnique (LMD), Paris, France

- Building neural surrogate models to accelerate computationally expensive atmospheric chemistry simulations, reducing inference time from hours to seconds for operational air quality prediction

Research Intern

Sept 2025 – Feb 2026

Danmark Meteorological Institute (DMI), Copenhagen, Denmark

- Lead author on a manuscript in preparation on AI-based predictive models for sea surface temperature and polar ice concentration forecasting using Multi-resolution architecture (4DVarNet)
- Exploration of generative AI approaches to quantify the uncertainties of the forecasts made

Leadership Experience

President, Télécom Paris Student Union | *Palaiseau, France*

April 2023 – June 2024

- Led a team of 26 to manage an annual budget of \$400,000 and all aspects of student life for 1,600 students
- Implemented initiatives to prevent alcohol-related risks, harassment, and sexual violence. Served as the primary liaison between the student body and school administration

Treasurer, Télécom Voile (Sailing Club) | *Perros-Guirec, France*

June 2023 – September 2024

- Managed an annual budget of €7,000 for the school's sailing association
- Leveraged 3 years of sailing instruction experience to enhance club activities

Notable Academic Projects

Additional projects at leguillouzic.com

SimCLR and BarlowTwins Implementation

2024

- IMAGE track project. Implementation of Google Deepmind's foundation paper of Self-Supervised Learning (Geoffrey Hinton et al.). Improvement with BarlowTwins (Yann Le cun et al.)

Cardiac Pathology Detection

2024

- Finished 6/75 of our internal school challenge (kaggle) for cardiac pathology detection using deep learning

Deepfake Detection using Convolutional Neural Networks

2023

- Generated synthetic faces using fine-tuned diffusion models (Stable Diffusion + LoRAs) and developed CNN-based detector (ResNet-18) achieving high accuracy on real vs. generated face classification

Technical Skills & Additional Informations

Areas of Expertise: Mathematics, Deep Learning, Image Processing, Signal Processing, Data Science

Languages: French (Native), English (Professional working proficiency), German (Elementary proficiency)

Interests: Geopolitics (actively follow international relations and policy developments), Competitive sports (participate in sailing competitions, french national running competitions and ski competitions)